

GABRIEL STEWART

Computer Science | AI, Software Development & Networked Systems
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PROFILE

Computer Science senior at Texas Tech University with experience spanning artificial intelligence, scientific computing, quantum computing, high-performance computing, and software development. Research member in the Texas Tech CPS Research Group, working with mathematical and graphical analysis of localized perturbations and dynamical-network stability. Builds Python-based simulations, data tools, and interactive applications, with additional experience in Linux, Git, JavaScript, SQL, C/C++, and technical communication.

EDUCATION

Texas Tech University — *Lubbock, TX* **Fall 2024–Present**
Bachelor of Science in Computer Science; senior-level student.
Academic areas include Data Structures and Algorithm Design, Artificial Intelligence, Quantum Computing, Programming Languages, and research on fragility in dynamical networks.

RESEARCH & PROFESSIONAL EXPERIENCE

Texas Tech CPS Research Group — *Research Member | Texas Tech University* **Aug 2025–Present**
Develop mathematical and graphical analyses to identify minimum localized perturbations and locations associated with destabilization in networked dynamical systems.
Collaborate in research meetings and presentations, communicating technical methods and results clearly to teammates.

Waxahachie ISD — *Technology Internship | Waxahachie, TX* **Summer 2026**
Contributed to the management of private content networks in a school-district technology environment.
Supported distribution and deployment of Dell hardware for district users.

Google Developers Club — *Member | Texas Tech University* **Jan 2025–Present**
Participate in technical and collaborative activities while adapting to new technologies and programming workflows.
Troubleshoot software issues and contribute to stable project development.

ADDITIONAL TECHNICAL EXPERIENCE

Quantum Computing — *Coursework & Project Experience | Lubbock, TX* **Jan 2025–May 2026**
Programmed with quantum-computing concepts and studied quantum hardware mechanics, including error-mitigation approaches and thermal considerations.
Built Qiskit-based logic for quantum gates, circuits, and state visualization in project work.

High-Performance Computing & Linux — *Technical Experience | Lubbock, TX* **Jan 2025–May 2026**
Used high-performance computing infrastructure and Linux environments to run technical workloads and manage compute-oriented workflows.

SELECTED PROJECTS

- Video-to-ASCII Converter** — *Python, OpenCV, API Development, CLI, Image Processing* [\[link\]](#) **May 2026–Present**
Developing a terminal-based application that converts video into ASCII art, with real-time/backlog processing and API integration for open-source media handling.
- Three-Body Problem Simulation** — *Python, MANIM, Physics Simulation, Data Visualization* [\[link\]](#) **Oct–Dec 2025**
Modeled and visualized three-body orbital dynamics with customizable simulation parameters and generated animation output.
- SQL/CSV & XLSX Data Analyzer** — *SQL, Data Analysis, Visualization*
Built a local application that analyzes CSV/XLSX datasets, identifies notable statistics and trends, and produces plots without cloud interaction.
- Discord Points & Leaderboard Bot** — *Cloud Services, Bot Development*
Created a Discord bot that tracks user points through cloud services and records a leaderboard to provide a consistent server-wide points system.
- Fluid Dynamics Image Transposition** — *Python, NumPy, PyQt5, Matplotlib* [\[link\]](#) **May–Aug 2023**
Applied fluid-dynamics concepts to image processing with real-time flow visualization, static-image controls, and computational modeling.
- Quantum Computing Game Logic** — *Qiskit, Matplotlib* [\[link\]](#) **Jan–Apr 2023**
Developed collaborative game logic for teaching quantum-computing fundamentals through simulated gates, circuits, and quantum-state visualization.
- Mathematical Cell Simulation** — *Python, argparse, multiprocessing* [\[link\]](#) **Sep–Dec 2022**
Implemented a state-transition simulation that determines each cell's next state from its current state and neighborhood, with multiprocessing support.

TECHNICAL SKILLS

Programming: Python, C, C++, Java, JavaScript, SQL, Lua, MATLAB

Tools & Libraries: Git, Linux, JupyterLab, OpenCV, MANIM, NumPy, Matplotlib, PyQt5, Qiskit, Microsoft Office

Technical Areas: AI / machine learning, scientific computing, data visualization, API development, command-line applications, frontend and backend development, high-performance computing

Professional Skills: Problem solving, leadership, conflict resolution, collaboration, technical presentation and public speaking

LEADERSHIP & SERVICE

- Capstone Opal Synthesis Project** — *Team Leader | Waxahachie, TX* **Aug 2023–May 2024**
Led a team through an opal-synthesis capstone, coordinating safe laboratory procedures and project execution.
Coordinated discussions on materials-handling and shipping regulations with German government officials.
- Global High School Esports Team** — *Vice President | Waxahachie, TX* **Jan 2023–May 2024**
Built stakeholder relationships, led team initiatives, and supported player development, teamwork, and competitive performance.
- League of Legends Esports Team** — *Team Leader | Waxahachie, TX* **Jan 2022–May 2024**
Set performance goals, delegated responsibilities by individual strengths, and resolved team conflicts to maintain collaboration.
- Felty Elementary School** — *Volunteer | Waxahachie, TX* **Aug 2023–May 2024**
Provided academic support and mentorship to elementary students in a collaborative learning environment.

CERTIFICATIONS & HONORS

Microsoft Word Certification
Microsoft Excel Certification
Graduated high school with an associate's degree
National African American Recognition Award
High School Mathematics Cord and Science Cord